

Linear Frequency Modulation Waveform Synthesis

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Abstract - Pulse compression plays an important role in design of the radar system. Pulse compression using linear frequency modulation techniques are very popular in modern radar. The linear frequency modulation is used to resolve two small targets that are located at long range with very small separation between them. The primary focus of this paper is the time frequency analysis and generation of LFM waveform using Direct Digital Chirp Synthesis (DDCS). This approach has been implemented on a Field Programmable Gate Array (FPGA) for the Synthetic Aperture Radar (SAR) application.

Keywords - Pulse compression, Linear Frequency Modulation, DDCS

I. INTRODUCTION

In modern pulsed Radar, range resolution (ΔR) is proportional to the pulse duration (τ). Therefore improved range resolution necessitates shorter pulse duration. Similarly the energy (E) content of the signal is also proportional to pulse duration (τ) and the detection probability depends it. Therefore to improve the detection, the pulse duration is required to be longer. To overcome this two conflicting requirements, pulse compression method is used [1]. The pulse compression usually done through Frequency Modulation and Phase Modulation are very popular in radars. Frequency modulation can be classified as Linear Frequency Modulation (LFM) and Nonlinear Frequency Modulation (NLFM).

LFM is the most popular radar waveform due to good range resolution and Doppler sensitivity. LFM waveform generation schemes are classified in analog and digital techniques. Analog pulse compression techniques are based on the surface acoustic wave (SAW) devices. However, design and fabrication of the SAW device for the large time-bandwidth product chirp signal is very complex and expensive, while the digital technique gives better advantage of programmability, flexibility, better stability, accuracy and repeatability [2]. This paper describes LFM generation and implementation in the Field Programmable Gate Array (FPGA).

II. LINEAR FREQUENCY MODULATION

In LFM, the frequency of the modulating signal increases linearly during the pulse duration of the signal [3]. In

a linear chirp, the time domain chirp signal is given by the equation 1.1.

$$s(t) = \exp(j\Phi(t)) \quad (1.1)$$

Where $\Phi(t)$ is the instantaneous phase, given by the equation as below:

$$\Phi(t) = 2\pi(f_0 t \pm Kt^2) \quad -\frac{T}{2} \leq t \leq \frac{T}{2} \quad (1.2)$$

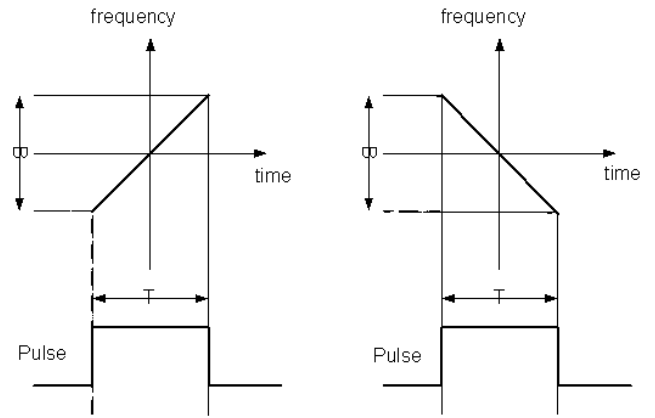


Figure 1- Typical LFM Waveforms (a) up-chirp (b) down-chirp

Where the f_0 is the center frequency (@ time $t = 0$) and K is the rate of the frequency increase or chirp rate.

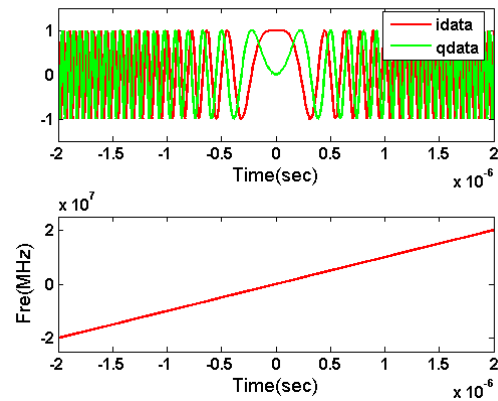


Figure 2 - Linear FM (a) I and Q data (b) Chirp rate

The LFM can be classified as ‘up chirp’ and ‘down chirp’. The up chirp and down chirp signal are given by the equation 1.3 and 1.4 respectively. Fig. 1 shows the freq vs time curve of typical LFM waveforms. The LFM waveform data are in complex form & contain In-phase (I-data) and quadrature-phase (Q-data) shown in Fig. 2. The pulse width is T, and the bandwidth is B.

$$\Phi(t) = 2\pi(f_0 t + K t^2) \quad (1.3)$$

$$\Phi(t) = 2\pi(f_0 t - K t^2) \quad (1.4)$$

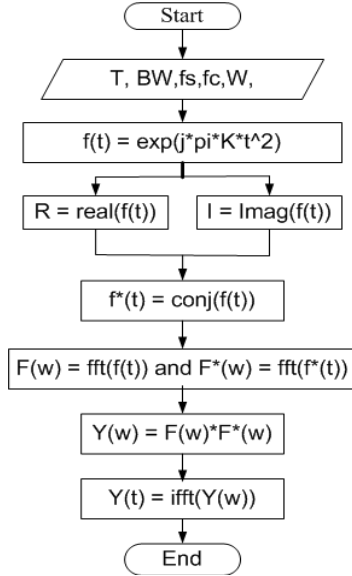


Figure 3 - LFM generation and matched filtering flowchart

The simulations related to LFM waveform generation is carried out using MATLAB. Figure 3 shows flowchart for matched filtering the LFM waveform generation and its range compression. The simulation results for the different conditions are discussed in the result and discussion section.

III. LFM HARDWARE IMPLEMENTATION

The digital LFM waveform generation is carried out using Direct Digital Chirp Synthesis (DDCS) technique [2]. In DDCS, the periodicity of sine wave is used as the limiting factor for memory storage. The block diagram of the DDCS based waveform generation is shown in Figure 4. The basic DDCS consists of the addition of a frequency accumulator block in front of the Direct Digital Synthesis logic (DDS) [4-6]. The implementation of the LFM waveform using DDCS technique is carried out in VHDL. The Transmit Pulse enables the DDCS only for that duration. The other inputs to the DDCS are the start frequency, delta frequency and start phase. Output of the DDCS consists of I-data and q-data that will drive through the output latch.

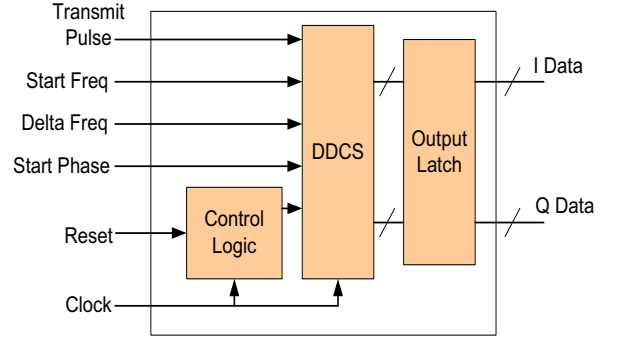


Figure 4 - DDCS for digital waveform generation

IV. RESULTS AND DISCUSSION

The simulation of LFM waveform is carried out for different specifications. The LFM signal is generated and correlated with the complex conjugate of same and the matched filter response are simulated. In all the cases, pulse duration is 20 μs. The simulations for the bandwidth variation from 100 MHz to 250 MHz are carried out. The first side lobe and the 3-db and 6-db width and resolution for different bandwidth options are shown in Table I. The correlation and the magnitude spectrum are shown in the Figure 5(a) and (b) respectively.

TABLE I
LFM RESULT FOR BANDWIDTH (250MHz TO 100MHz)

Sr. No.	BW (MHz)	-3db Δτ (nsec)	-3db ΔR = cΔτ/2 (m)	-6db Δτ (nsec)	-6db ΔR = cΔτ/2 (m)	1 st Side lobe (db)
1	250	1.8	0.27	2.4	0.36	13.26
2	225	1.93	0.4	2.66	0.72	13.26
3	200	2.2	0.33	3.0	0.45	13.26
4	175	2.53	0.38	3.46	0.52	13.26
5	150	2.93	0.44	4.0	0.6	13.26
6	125	3.6	0.54	4.8	0.72	13.26
7	100	4.4	0.66	6.0	0.9	13.26

ΔR = range resolution, Δτ = compressed width

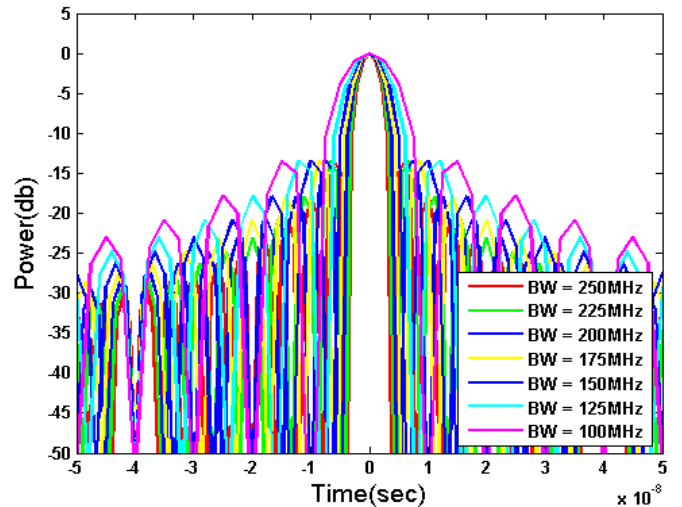


Figure 5 (a) Correlation for Different BW

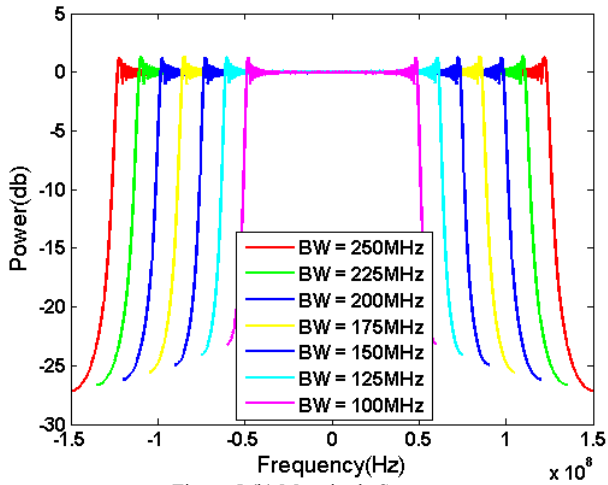


Figure 5 (b) Magnitude Spectrum

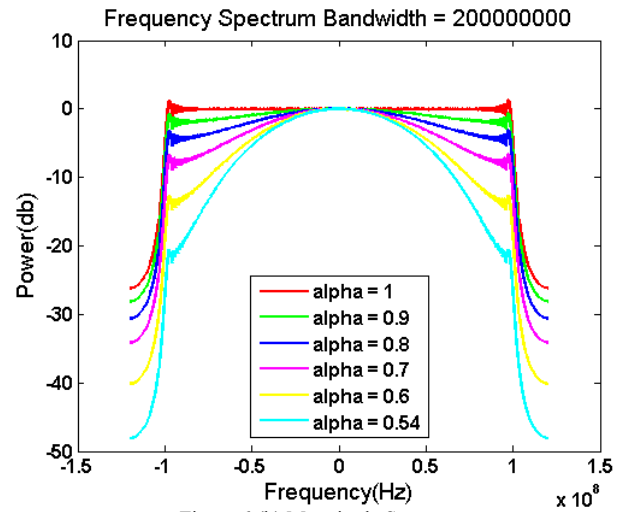


Figure 6 (b) Magnitude Spectrum

Similarly simulation for the weighted LFM waveform is carried out using hamming window for signal bandwidth of 200MHz. The basic hamming window equation is given by equation 1.5. The simulations for weighting factor variation from 0.54 to 1 are carried out. The first and second side lobe and the 3-db width and resolution for different weighting factor (alpha) are shown in Table II. The correlation and the magnitude spectrum are shown in the Figure 6(a) and (b) respectively.

$$Y(f) = \alpha + (1 - \alpha) \cos(2\pi \frac{f}{B}) \quad (1.5)$$

TABLE II
LFM RESULT HAMMING WEIGHT (0.54 TO 1)

Sr. No.	Hamming Weight Factor (alpha)	-3db $\Delta\tau$ (E-9)	-3db $\Delta R = c\Delta\tau/2$ (m)	Side lobe Peak(db)	
				1 st	2 nd
1	1	2.20	0.33	-13.26	-17.83
2	0.9	2.26	0.34	-15.34	-19.07
3	0.8	2.40	0.36	-18.64	-20.92
4	0.7	2.60	0.39	-24.99	-24.08
5	0.6	2.93	0.44	-46.62	-31.59
6	0.54	3.26	0.49	-43.97	-55.60

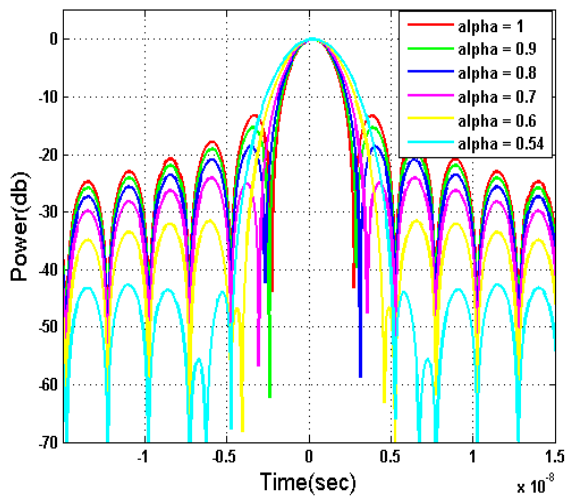


Figure 6 (a) Correlation for different weight

The LFM waveform generation using DDCS has been simulated using Modelsim software. Figure 7 shows the simulation result in time domain. The simulation is carried out for the generation of the signal having pulse width of 20us. The DDS clock rate is 125MHz, Spurious Frequency Dynamic Range (SFDR) is 60 dB and frequency resolution is 500 Hz.

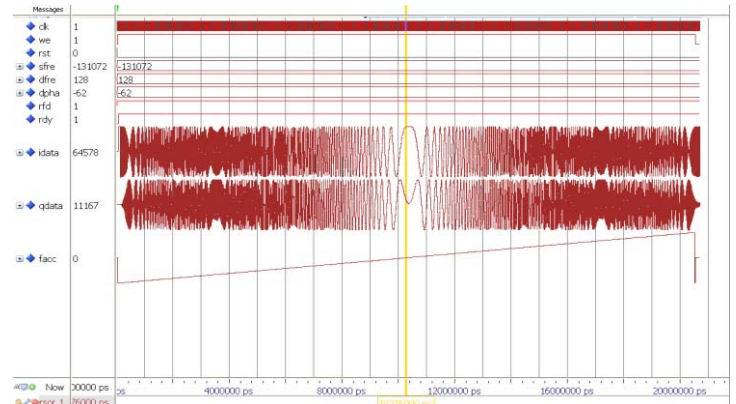


Figure 7 Simulation result for linear FM waveform using DDCS

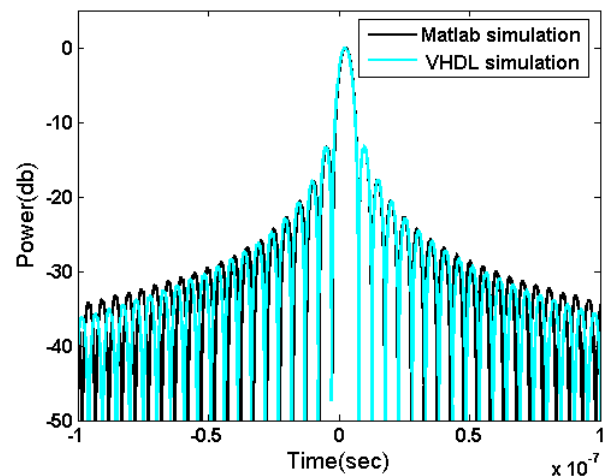


Figure 8 (a) MATALB and Modelsim Simulation comparison

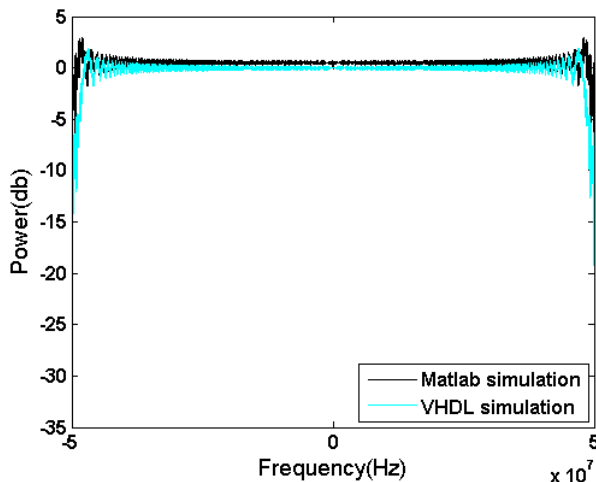


Figure 8 (b) MATALB and Modelsim Simulation comparisons

The DDS output width is 10 bits and data width and accumulator width are 18 bits. The phase angle width is 15 bits and latency is 3 clock cycles. The comparison between the MATLAB simulation and the LFM generated through Modelsim simulation are shown in Figure 8(a) and 8(b).

V. CONCLUSION

Pulse compression using LFM technologies is a useful technique for SAR; it is an enabling technology to facilitate the use of the low power component in the transmitter. It also has the benefit of improving the dynamic range and range resolution of the radar. LFM generation using DDCC techniques has lower computational complexity

compared to other techniques and better programmability, flexibility and repeatability.

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